Beyond RAG: Domain-Specific Agentic Architecture for Biomedical Knowledge Graph Construction

Umesh Bhatt • umesh@8thcross.com • March 2026

From GraphRAG to Epistract — evolving latent knowledge graphs for drug discovery

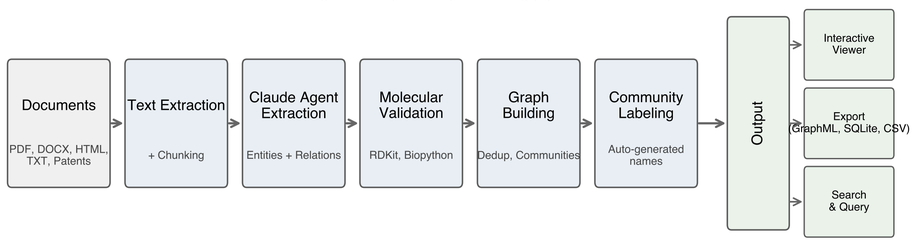
Motivation

Knowledge graphs are essential for drug discovery, yet traditional construction requires **domain-specific models retrained per therapeutic area**. GraphRAG showed LLMs can extract knowledge without training, but produces **untyped, similarity-based edges** — a scientist cannot distinguish a side effect from a mechanism of action.

Epistract replaces proximity with **comprehension**: parallel LLM agents extract typed relations against a domain schema and validate molecular identifiers with deterministic tools.

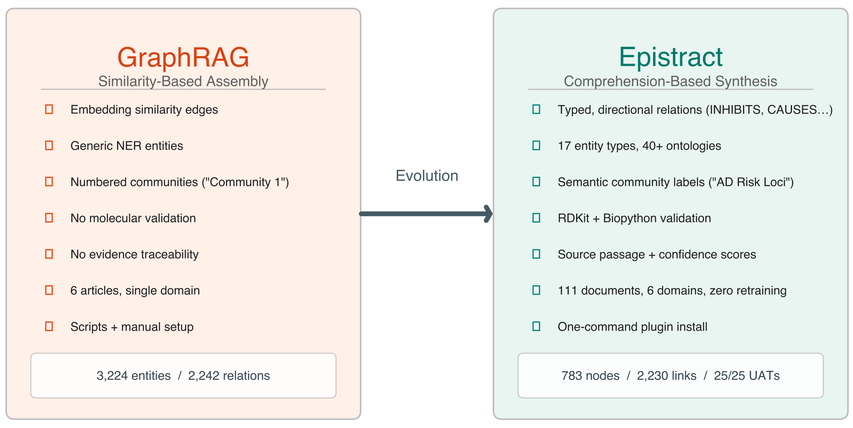
Architecture

Figure 1. Epistract processing pipeline



1. **Parallel extraction** — one LLM agent per document
2. **Schema enforcement** — 17 entity types, 30 relation types
3. **Molecular validation** — RDKit, Biopython
4. **Graph construction** — dedup, communities, semantic labels

Comprehension vs. Co-occurrence



Domain Schema

17 Entity Types in 6 categories:

Drug & Chemistry	2
Molecular Biology	5
Disease & Phenotype	3
Clinical	3
Pathways & Mechanisms	2
Context	2

30 Directed Relation Types:
Drug–Target (5), Drug–Disease (2), Drug–Clinical (2), Drug–Drug (3), Molecular Biology (12), Biomarker (2), Organizational (3), Fallback (1)

Grounded in **40+ ontologies**:
ChEBI, HGNC, UniProt, MeSH, MedDRA, KEGG, Reactome

Evaluation: Six Drug Discovery Domains

#	Domain	Docs	Nodes	Links	UATs
1	Neurogenetics (PICALM / Alzheimer's)	15	149	457	4/4
2	Oncology (KRAS G12C inhibitors)	16	108	307	5/5
3	Rare Disease (PKU, Achondro., NPC)	15	94	229	3/3
4	Immuno-Oncology (Checkpoints)	16	132	361	4/4
5	CV / Inflammation	15	94	246	3/3
6	GLP-1 Competitive Intel.†	34	206	630	6/6
	Total	**111**	**783**	**2,230**	**25/25**

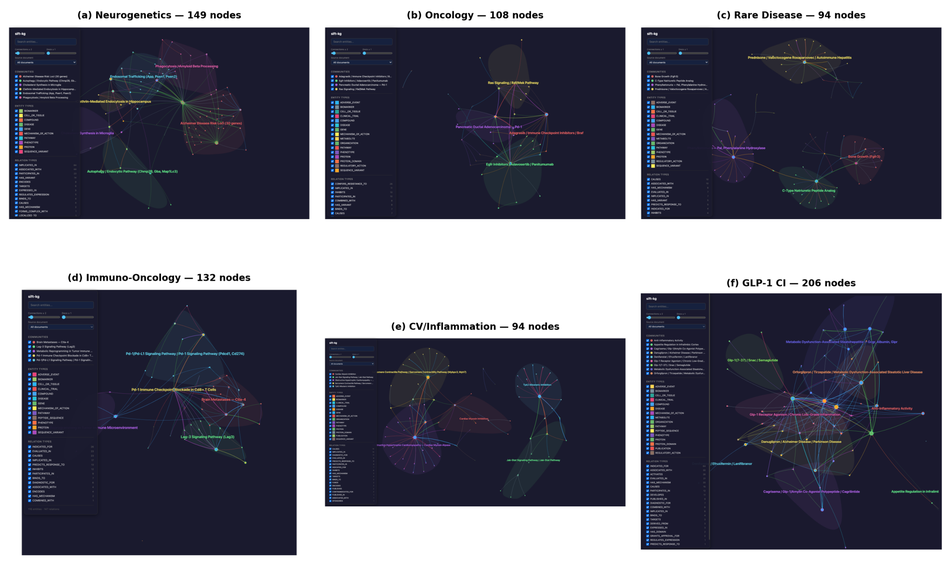
†Multi-source: PubMed + Google Scholar + Google Patents via SerpAPI

100% Entity Type Coverage

93% Relation Type Coverage

25/25 UATs Passed

Knowledge Graph Visualizations



33 auto-labeled communities across 6 scenarios. Community detection correctly separates unrelated therapeutic areas and identifies commercially meaningful clusters.

Key Contributions

- **Method:** Agentic extraction with typed domain schemas, parallel agent dispatch, and molecular validation (RDKit + Biopython)
- **Tool:** Open-source Claude Code plugin, installable in one command, with all test corpora and outputs committed for reproducibility
- **Multi-source:** PubMed + Google Scholar + Google Patents — mirrors how human researchers conduct competitive intelligence

Open Source • MIT License • github.com/usathyan/epistract

Install: `claude plugin install epistract`